💡 Conductivity Lab: What Lets Electricity Through?: Lab Sheet · ANSWER KEY

Chemistry · Unit 1C · Level 3 · goes with L1.13 No Swimming, L1.13.5 You Light Up My Life

Name: _____

Question: How does dissolving a substance in water affect whether it conducts electricity?

Materials

conductivity tester (low voltage), foil or copper, wax, sugar, salt, distilled water, cups, balance and graduated cylinder

⚠️ Safety

- Goggles on the whole lab.
- Never taste anything, even salt or sugar.
- Only the low-voltage tester. Nothing in wall outlets.
- Wash hands after.

Prediction

Predict which materials will conduct. Explain using electrons or ions.

Metal (free electrons) and salt water (ions) will conduct; wax, sugar, solid salt and pure water will not (or only very little).

Procedure

1. Check that the tester bulb lights when its two ends touch.
2. Test each solid: foil/copper, wax, sugar, salt crystals. Record.
3. Make 100 mL salt water at 1%, 5% and 10% (1 g, 5 g, 10 g of salt). Test each.
4. Test distilled water and sugar water.
5. Dry the tester between every test.
6. Sort the materials into conductors and insulators.

Data

Material	Prediction	Bulb: bright / dim / off	Conductor or insulator?
foil / copper		**bright**	**conductor**
wax		**off**	**insulator**
salt crystals		**off**	**insulator**
distilled water		**off**	**insulator**
1% salt water		**dim**	**conductor**
10% salt water		**bright**	**conductor**
sugar water		**off**	**insulator**

Expected results (your class data may differ).

Analysis (Claim, Evidence, Reasoning)

1. Claim: How does dissolving salt in water affect conductivity?
Dissolving salt makes water conduct; more salt conducts more.

2. Evidence: Use your data for distilled water, 1% and 10% salt water.
Distilled water: off; 1%: dim; 10%: bright.

3. Reasoning: Why does sugar water NOT conduct? Use the word ions.
Sugar dissolves as neutral molecules, not ions, so nothing charged can move to carry current.

4. **Current to charges: 1 A = 6.28 × 10^{18} charges per second.**

> Example: 0.2 A = 0.2 × 6.28 × 10^{18} = 1.256 × 10^{18} charges per second.

a) 0.5 A = ____ × 10^{18} charges/s **3.14**

b) Which moves more charge each second: 0.1 A or 0.4 A? **0.4 A (4 times as much)**

🔑 *Key written by our team from Mr. Brittle's lab sheets, please check.*